METRIC

MIL-PRF-85045/21A(SH)
5 November 2021

SUPERSEDING
MIL-PRF-85045/21(SH)
W/AMENDMENT 1
18 April 2017

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, EIGHT FIBERS, ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 2 (OFCC), FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), CABLE CLASS SM AND MM

This specification is approved for use by Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type: 2 (optical fiber cable component [OFCC]).

Fiber Cable Class: MM (graded-index, glass core and glass cladding, multimode)

SM (dispersion-unshifted, glass core and glass cladding, single-mode)

DESIGN AND CONSTRUCTION:

Fiber:

Class MM fibers shall be in accordance with MIL-PRF-49291/6.

Class SM fibers shall be in accordance with MIL-PRF-49291/7.

Buffer diameter: 900±50 micrometers (µm).

OFCC:

Dimensions and configuration: See figure 1.

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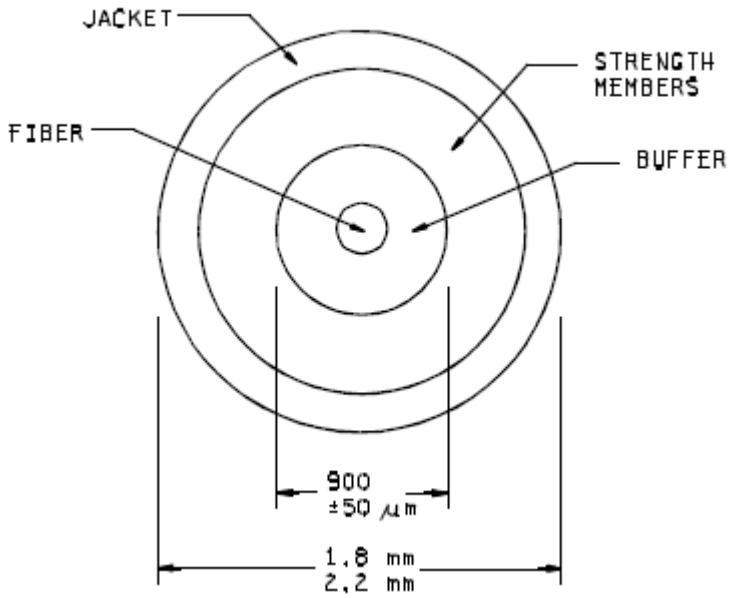


FIGURE 1. OFCC.

Mass per unit length: ≤15 kilograms per kilometer (kg/km).

Tensile loading: ≥270 newtons (N).

Dynamic bend tensile load: 90 N minimum.

Jacket material: The OFCC jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Short term minimum bend diameter: Eight times the OFCC outer diameter.

Long term minimum bend diameter: Sixteen times the OFCC outer diameter.

Finished cable:

Dimensions and configuration: See figure 2. Eight OFCC units shall be helically laid over the central member. Materials with acoustic performance properties similar to those of Mylar shall not be used. The outer jacket may be composed of multiple layers. The total jacket wall thickness shall be a minimum of 2.8 millimeters (mm) (establishes compatibility with termination and penetration devices).

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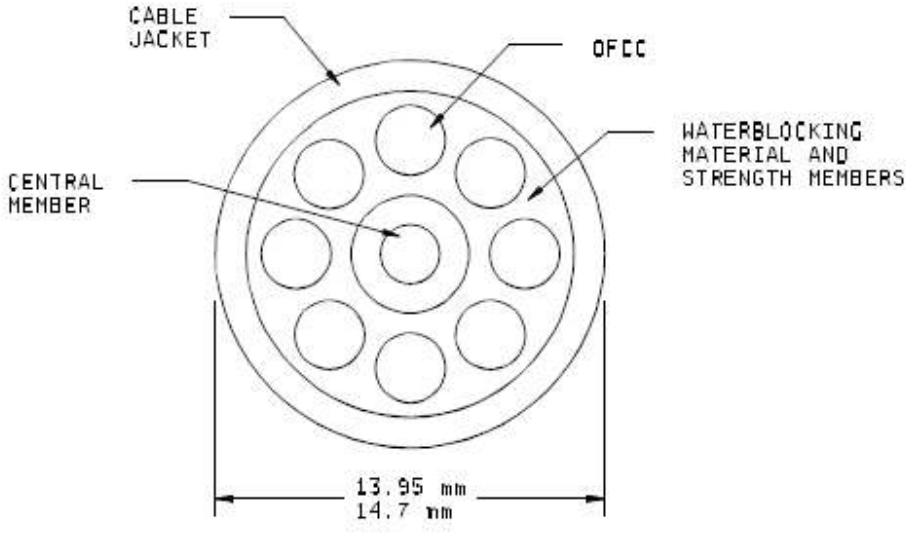


FIGURE 2. Eight-OFCC fiber optic cable.

Number of fibers: Eight (one per OFCC).

Concentricity: ≥0.65.

Mass per unit length: ≤215 kg/km.

Jacket material: The outer jacket shall be arctic type polychloroprene (black).

Short term minimum bend diameter: Eight times the cable outer diameter. (The short term minimum bend diameter shall be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long term minimum bend diameter: Sixteen times the cable outer diameter.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

PERFORMANCE REQUIREMENTS: The individual performance requirements shall be as specified in MIL-PRF-85045 and this specification sheet, including the requirements identified in table I.

Optical Properties:

Maximum attenuation rate: 4.5 decibels (dB)/km at 850±20 nanometers (nm), 2.0 dB/km at 1300±20 nm for class MM fiber.

2.0 dB/km at 1,310±20 nm and 1,550±20 nm for class SM fiber.

Bandwidth: Fiber with a minimum bandwidth of 500 megahertz (MHz)-km at 1,300 nm and 160 MHz-km at 850 nm shall be used (multimode cables only).

Change in optical transmittance: Measurements shall be made at 1,300±20 nm.

Mechanical properties:

Tensile loading and elongation: Applicable; tensile loading ≥2,700 N.

Low temperature flexibility (cold bend): The exposure temperature shall be -40 °C.

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Cyclic flexing: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same test position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable twist bending: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same test position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Pressure cycling: Applicable. The change in optical transmittance shall be less than 1.0 dB during the test and 0.5 dB after the test.

Hosing: Both low pressure and hydrostatic pressure are applicable.

Low pressure: The test pressure shall be 379 kilopascals (kPa). Leakage shall be not greater than 400 milliliters (mL) after 2 hours and not greater than 1,000 mL after 6 hours.

Hydrostatic: The test pressure shall be 7.7 MPa. There shall be no leakage through the cable core during the first 2 hours. The total leakage (cable core and OFCC) shall be not greater than 3,000 mL after 6 hours.

Cable scraping resistance: 750 cycles.

Cable to cable abrasion: 500 cycles.

Environmental properties:

Temperature range:

Operating:	-28 to 65 °C
Nonoperating:	-40 to 70 °C
Storage:	-40 to 70 °C

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Temperature humidity cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Life aging: The jacket material shall be tested at 100 °C for 168 hours. Tensile strength and elongation retention shall be not less than 50 percent.

Fluid immersion: Exposure to automobile gasoline and tap water are not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C), and lubricating oil (98 to 100 °C).

Shock: Applicable. External wireway mounting shall be used.

Water absorption: 10 milligrams per square centimeter maximum.

Chemical properties:

Acid gas generation: Applicable, except that only OFCCs removed from specimen will be tested. OFCCs will be cut to approximately 2.5 centimeters long and 16 OFCCs will be required to achieve mass of minimum 1.25 grams to maximum 2.5 grams and are to be placed in silica boat for test.

Halogen content: <0.2 percent

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Cross-link verification: This test is applicable for the cable jacket only. The test shall be conducted in accordance with ICEA T-28-562 and run at 200 °C. The test shall be sequenced after the weathering test in the qualification test sequence and after the fluid immersion test in the group C conformance test sequences. The hot creep shall not exceed 100 percent and the hot creep set shall not exceed 10 percent.

Cable Jacket Tear Strength: the cable jacket tear strength shall be not less than 40 Newtons per centimeter (N/cm) of jacket thickness.

Part or identifying number (PIN):

M85045/21-01 (Multimode).

M85045/21-02 (Single-mode).

Qualification and conformance inspection: See table I. This table specifies applicable requirements, qualification inspections, and conformance inspections per this specification sheet. The Requirement Paragraph and Test Paragraph columns within this table reference MIL-PRF-85045. The Cable Length column of this table applies only to the qualification inspection. A reduced sample set is allowed for conformance inspection; refer to MIL-PRF-85045 for conformance inspection sample size.

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TABLE I. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each 4/	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each 5/	A
II	Crosstalk	3.5.3	4.7.4.3	Three samples, 0.5 km each 5/	--
III	Temperature cycling	3.7.1	4.7.6.1	Two samples, 0.5 km each 5/ (one on reel, one off reel)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each 6/	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each 6/	--
	Low temperature flexibility (cold bend)	3.6.1	4.7.5.3	Two specimens, 8 m each 7/	B
	Cyclic flexing	3.6.4	4.7.5.4	Six specimens, 5 m each 8/ (two specimens at each temperature)	--
	Crush	3.6.5	4.7.5.5	Two specimens, 5 m each 7/	--
	Cable twist-bending	3.6.6	4.7.5.6	Six specimens, 5 m each 8/ (two specimens for each temperature)	C
	Impact	3.6.8	4.7.5.8	Two specimens, 5 m each 7/	C
	Temperature life (life aging)	3.7.6	4.7.6.6	Two specimens, 300 m each 7/	C
	Tensile loading and elongation	3.6.1	4.7.5.1	Two specimens, 150 m each 9/	C
	Operating tensile load	3.6.1.1	4.7.5.1.1	Two specimens, 150 m each 10/	C
	Freezing water immersion (ice crush)	3.7.7	4.7.6.7	Two specimens, 30 m each 7/	--
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each 7/	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each 7/	C

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/, 2/, 3/]	Conformance inspection
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.49 km [5/] (on reel)	--
	Weathering	3.7.8	4.7.6.8	One specimen, 2 m [11/] and three material samples [12/]	--
	Fluid immersion	3.7.9	4.7.6.9	One specimen, 2 m [11/] and three material samples [12/] for each specified fluid	C
	Paint susceptibility	3.7.15	4.7.6.15	Two specimens, 2 m each [13/]	--
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m [11/]	--
	Shock	3.7.13	4.7.6.13	One specimen, 30 m [11/]	--
	Dynamic bend	3.6.2	4.7.5.2	One specimen, 150 m [14/]	B
	Hosing: Low pressure	3.6.12.1	4.7.5.12.1	One specimen, 1.5 m [11/]	C
	Hosing: Hydrostatic	3.6.12.2	4.7.5.12.2	One specimen, 2 m [11/]	C
	Radial compression (for application B only)	3.6.7	4.7.5.7	One specimen, 10 m [11/]	--
	Pressure cycling (for application B only)	3.6.11	4.7.5.11	One specimen, 30 m [11/]	C
	Corner bend	3.6.9	4.7.5.9	Two specimens, 5 m each [13/]	--
	OFCC kink	3.4.2.2.2	4.7.3.5	Two specimens, 1 m each [11/]	--
	Cross-link verification	[15/]	[15/]	Two specimens, 1 m each [13/]	C
V	Dripping	3.6.13	4.7.5.13	One specimen, 30 cm [11/]	--
	Cable jacket tear strength	3.6.14	4.7.5.14	Three specimens, 1 m each [13/]	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	Five specimens [16/]	--
	Cable abrasion resistance	3.6.16	4.7.5.16	Four specimens, 2 m each [17/]	C
	Cable shrinkage	3.6.17	4.7.5.17	Three specimens, 0.5 m each [13/]	C
	Durability of identification marking	3.6.19	4.7.5.19	Three specimens, 2 m each [13/]	--
	Water absorption	3.7.14	4.7.6.14	Two specimens [18/]	--

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/	Conformance inspection
VI	Acid gas generation (OFCC only)	3.8.1	4.8.1	One specimen, 1 m 19/	C
	Halogen content (OFCC only)	3.8.2	4.8.2	One specimen, 1 m 19/	--
	Toxicity index (OFCC only)	3.8.3	4.8.3	One specimen, 1 m 19/	C
	Smoke index (OFCC only)	3.8.5	4.8.6	One specimen, 1 m 19/	C

NOTES:

1/ Tolerance on 0.5 kilometer length is ±10 percent provided results are normalized to 1 kilometer.

2/ Tolerance on shorter lengths is ±10 percent.

3/ Mixed fiber cables, consisting of alternating single-mode and multimode fibers, can be used for qualification purposes. If a failure occurs for any fiber in a mixed fiber cable, then all of the fibers in the cable are considered to have failed (both single-mode and multimode).

4/ The visual and mechanical inspection shall only be conducted on a 2-meter section of each sample.

5/ The same samples as used in the visual and mechanical inspection shall be used.

6/ The same samples as used in the temperature cycling test shall be used: one on reel, one off reel.

7/ A specimen cut from each sample used in the temperature cycling test shall be used.

8/ Three specimens cut from each sample used in the temperature cycling test shall be used.

9/ A specimen cut from each sample used in the life test shall be used.

10/ The same sample as used in the tensile loading and elongation test shall be used.

11/ A specimen cut from the specimen used in the thermal shock test shall be used.

12/ Three jacket material samples as specified in 4.7.5.15 shall be used in this test.

13/ Specimens cut from the specimen used in the thermal shock test shall be used.

14/ A specimen cut from the specimen used in the thermal shock test or one of the specimens used in the tensile loading and elongation test shall be used.

15/ As stated under the applicable tests in this specification sheet.

16/ Jacket material samples as specified in 4.7.5.15 shall be used in this test.

17/ Specimens cut from the specimen used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion testing and two specimens shall be used in the cable-to-cable abrasion testing.

18/ Jacket material samples as specified in 4.7.5.15 shall be used.

19/ A specimen of OFCC cut from one of the samples used in the temperature cycling or thermal shock tests shall be used.

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Qualification and conformance by similarity - fiber type: Manufacturers who are qualified under this specification sheet for multimode (single-mode) fiber cable and whose single-mode (multimode) fiber cable passes the visual and mechanical, attenuation rate, temperature cycling, temperature/humidity cycling, storage temperature, cyclic flexing, crush, cable twist-bending, impact (low temperature only), tensile loading and elongation, operating tensile loading, thermal shock, and dynamic bend inspections specified herein, are qualified under this specification sheet for single-mode (multimode) fiber cable. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from a multimode (single-mode) fiber to a single-mode (multimode) fiber. Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

TABLE II. Qualification and conformance by similarity (fiber).

Group	Inspection	Qualification inspection 1/	Conformance inspection 2/
I	Visual and mechanical inspection	X	A
	Attenuation rate	X	A
II	Temperature cycling	X	C
	Temperature humidity cycling	X	C
	Storage temperature	X	--
III	Cyclic flexing	X	--
	Crush	X	--
	Cable twist-bending	X	C
	Impact (low temperature only)	X	C
	Tensile loading and elongation	X	C
	Operating tensile load	X	C
IV	Thermal shock	X	--
	Dynamic bend	X	B

NOTES:

1/ Where manufacturers are qualified for M85045/21-01 (M85045/21-02) and are pursuing qualification for M85045/21-02 (M85045/21-01).

2/ Where manufacturers are qualified for M85045/21-01 and M85045/21-02 and are performing conformance testing for M85045/21-02 (M85045/21-01).

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 6015-2020-006)

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